

An Asymmetry in Havlík’s Law: Evidence That Liquid-Context Jers Were Not Real Vowels

Marko Jevremović

Independent researcher, Belgrade, Serbia

`marko@asemic.me`

Abstract

This paper documents a categorical asymmetry in the outcomes of Havlík’s law. In Czech, strong-position jers in non-liquid environments vocalize to /e/ in 100% of cases ($n = 23$), while strong-position jers adjacent to liquids vocalize in 0% of cases ($n = 27$), yielding syllabic consonants instead. The same split is replicated independently in Serbian ($n = 22$) and Slovak ($n = 23$): across three branches, 68/68 non-liquid strong jers vocalize while 0/27 liquid-adjacent ones do. The seven Modern Czech words with /e/ near liquids are shown to be secondary epenthesis (13th–15th c.), post-dating the jer fall by centuries. Cross-Slavic evidence from Russian (polnoglasie-like vowel quality at liquid positions) and Bulgarian (front jer yielding back vowel before liquids) corroborates the pattern. We argue that liquid-context jers were notational conventions for syllabic consonants, not real reduced vowels, and that Havlík’s law applies only to genuine reduced vowels from PIE **i/ *u*.

1 Introduction

The fall of the jers (падение еров) is among the most consequential sound changes in Slavic historical phonology (Townsend & Janda 1996; Matasović 2014). Proto-Slavic had two reduced vowels, the back jer ъ and the front jer Ѣ, which by the 10th–12th centuries were either strengthened to full vowels or lost entirely, depending on their position in the word. The governing regularity is Havlík’s law: counting from the end of the word, jers in odd-numbered positions (“weak”) are deleted, while jers in even-numbered positions (“strong”) vocalize to full vowels—typically /e/ in Czech and Slovak, /o/ or /e/ in Russian (see Havlík 1889; Shevelov 1964; Bethin 1998).

The law is remarkably regular for jers in non-liquid environments. Proto-Slavic **sъnъ* ‘sleep’ yields Czech *sen*, **dъnъ* ‘day’ yields *den*, **pъsъ* ‘dog’ yields *pes*—in each case the strong jer vocalizes to /e/ while the weak jer is deleted.

But for jers adjacent to liquids (/r/, /l/), the outcome is categorically different. Proto-Slavic **vьlkъ* ‘wolf’ yields Czech *vlk*—not **velk*. **sъmьrtъ* ‘death’ yields *smrt*—

not **smert*. **pǫrstǫ* ‘finger’ yields *prst*—not **perst*. The strong-position jer does not vocalize to /e/; instead, it disappears, and the liquid becomes a syllabic consonant.

This asymmetry is well known descriptively: any Czech grammar notes that the language has syllabic /r/ and /l/. What has not been explicitly documented, to the best of our knowledge, is the **categorical nature** of the split and its implications for the phonological status of liquid-context jers. In this paper we present systematic data showing that the asymmetry is **total**—100% vocalization for non-liquid jers versus 0% for liquid-adjacent jers—and argue that this categorical split constitutes evidence for a fundamental distinction between two types of graphemes written identically as ʁ/ʁ in OCS:

1. **Real reduced vowels** (from PIE short **u*, **i*), which participate in Havlík’s law normally;
2. **Notational conventions for syllabic consonants** (from PIE syllabic resonants **l̥*, **r̥*), which were never real vowels and therefore have nothing to vocalize.

This reanalysis eliminates the need for an ad hoc exception in Havlík’s law for liquid contexts and simplifies the diachronic account of Slavic syllabic consonants.

2 Background

2.1 Havlík’s law

Havlík’s law (Havlík 1889) is typically formulated as follows. In a sequence of jers counted from the right edge of the word, the rightmost jer is weak (position 1), the next is strong (position 2), and so on, alternating. A full vowel resets the count. Weak jers are deleted; strong jers are vocalized to full vowels.¹

The law operates with near-exceptionless regularity across Slavic for jers in non-liquid environments (see §3.1 below). Its status as a genuine phonological regularity rather than a mere descriptive generalization is supported by modern experimental evidence: Gouskova & Becker (2013) showed that Russian speakers productively apply yer-like alternations to nonce words, with special constraints near sonorants.

2.2 PIE syllabic resonants and their Slavic reflexes

Proto-Indo-European had four sonorant consonants—**m*, **n*, **l*, **r*—that could function as syllable nuclei in zero-grade ablaut environments, written **m̥*, **n̥*, **l̥*, **r̥* (see Beekes 2011; Fortson 2010). Each IE daughter branch resolved these syllabic resonants

¹The specific reflex varies by branch: /e/ in Czech and Slovak, /o/ (for ʁ) or /e/ (for ʁ) in Russian, /a/ in Serbian, etc.

differently, inserting a distinct epenthetic vowel: Sanskrit preserved the syllabic resonant (*ṛ*), Greek inserted /a/, Latin /o/, Germanic /u/, Lithuanian /i/, and Slavic wrote jer + liquid (*ѣr*, *ѣl*, etc.; Lunt 2001).² The standard reconstruction posits a Proto-Balto-Slavic intermediate stage **iR*/**uR* (Mayer 1991), with ~73% showing **i* (reflected as front jer *ѣ*).

The question at issue is whether this intermediate PBS vowel was a “real” full short vowel that later reduced to a jer (the standard view), or whether Slavic preserved the syllabic resonant more or less directly from PIE, with the jer serving as a scribal device to represent the syllabic function of the consonant (the alternative we develop here).

2.3 Previous treatments of liquid-context jers

Scheer (2003, 2004) provides the most detailed structural analysis of Czech syllabic consonants within a Government Phonology (CVCV) framework. He distinguishes **syllabic consonants** (from Common Slavic **tbrt*, where the jer precedes the liquid) and **trapped consonants** (from CS **trbt*, where the jer follows). In his framework, the jer is lost and the liquid spreads into the vacated nuclear position. Scheer does not frame the *vlk*/*sen* difference as a problem for Havlík’s law; in his framework, both outcomes are natural structural consequences of sonorant spreading.

Shevelov (1964) argued that Common Slavic jers were not a special “reduced” category but simply short vowels—the short members of the vowel system. This opens the theoretical space for liquid-context jers to have had a different phonetic realization (minimal or zero vocalic content) from non-liquid jers, even if they were written identically.

Birnbaum (1976) examined Slavic liquid diphthongs and their resolution across branches, noting the diversity of outcomes but without framing it as a Havlík’s law problem. Shafirovich (2016) investigated “the possibility of existence of syllabic liquids in Common Slavic” by examining variation between *ѣr*/*t* and *rѣt* spellings in manuscripts, sometimes appearing in the same word without semantic distinction. This is the closest prior work to our argument, but it did not assemble the quantitative case we present here.

Isačenko (1970) proposed a revision of Havlík’s law for East Slavic, arguing that the jer fall in Russian was governed by morphophonemic rather than purely phonological conditions. His revision focused on the interaction between jer vocalization and morphological boundaries, but did not isolate the liquid/non-liquid split as a categorical distinction. The present paper differs in identifying the asymmetry as a *phonological* fact about liquid adjacency, observable most cleanly in Czech, Serbian, and Slovak

²The nasal syllabic resonants **m̥* and **n̥* followed a different path in Slavic, yielding nasal vowels (ѧ, ѧ̃) rather than jer + nasal sequences. They are therefore irrelevant to the Havlík’s law asymmetry discussed here.

where the outcomes are transparent.

Bethin (1998) treats the liquid+jer interaction through the lens of prosodic structure, arguing that syllable reorganization preceded the fall of the jers and that liquid-adjacent jers occupied a distinct prosodic position. In a related vein, Yoo (2012), building on Bethin’s (1992, 1993) autosegmental analysis of Late Common Slavic syllable structure, argues that the “tense jers” before jod in extreme eastern East Slavic dialects were not genuine jer vowels at all but reflect a different syllabification of the **ǔ/ *ĩ*+jod sequence—a conclusion structurally parallel to ours for a different phonological environment.

Notably, **none of these works explicitly identifies the 100% vs. 0% vocalization asymmetry** documented in the present study, nor draws the inference that liquid-context jers may have been non-vocalic.

3 The Czech data

Czech is the ideal primary test language for this investigation because it preserves both syllabic /r/ and /l/ as productive phonological categories, making the liquid-adjacent jer outcomes maximally transparent.

3.1 Category A: Strong jers not adjacent to liquids

Table 1 presents all identified cases where a strong-position jer in a non-liquid environment is attested in Proto-Slavic and has reflexes in Czech, Serbian, and Slovak—the three languages whose outcomes are most transparent for this investigation.³

The result is unambiguous: every strong-position jer in a non-liquid context vocalizes to a full vowel—/e/ in Czech and Slovak, /a/ in Serbian—across all three languages. The vocalization rate is **100%** in each branch independently (Czech $n = 23$; Serbian $n = 22$, excluding one lost root; Slovak $n = 23$). Weak-position jers in the same words (**sǣnǫ*, **dǫnǫ*) delete as predicted, confirming the standard strong/weak alternation.

Methodological note: The corpus was assembled by systematically extracting Proto-Slavic reconstructions containing jers from Derksen (2008), supplemented by Carlton (1991) and Shevelov (1964), and filtering for those with strong-position jers in non-liquid environments and transparent modern reflexes. The relevant statistical point is not the sample size but the **categorical nature** of the split—68/68 vocalized jers across three independent branches vs. 0/27 in liquid-adjacent environments—which no reasonable alternative model predicts. Even if additional examples exist, a single

³Czech and Slovak preserve both syllabic /r/ and /l/ (Kenstowicz & Rubach 1987), making the liquid-adjacent contrast maximally visible. Serbian preserves syllabic /r/ and resolved syllabic /l/ to /u/, providing an independent test: if the jer had vocalized by Havlík’s law, the outcome should be /a/ (the regular Serbian jer reflex), not /u/.

Table 1: Strong-position jers in non-liquid environments: cross-Slavic outcomes.

#	Proto-Slavic	Gloss	Czech	Serbian	Slovak
1	* <i>sonō</i>	sleep	<i>sen</i>	<i>san</i>	<i>sen</i>
2	* <i>donō</i>	day	<i>den</i>	<i>dan</i>	<i>deň</i>
3	* <i>posō</i>	dog	<i>pes</i>	<i>pas</i>	<i>pes</i>
4	* <i>moxō</i>	moss	<i>mech</i>	<i>mah</i>	<i>mach</i>
5	* <i>lonō</i>	flax	<i>len</i>	<i>lan</i>	<i>lan</i>
6	* <i>vosŕō</i>	louse	<i>veš</i>	<i>vaš</i>	<i>voš</i>
7	* <i>otōcō</i>	father	<i>otec</i>	<i>otac</i>	<i>otec</i>
8	* <i>konōcō</i>	end	<i>konec</i>	<i>konac</i>	<i>koniec</i>
9	* <i>kupōcō</i>	merchant	<i>kupec</i>	<i>kupac</i>	<i>kupec</i>
10	* <i>švō</i>	seam	<i>šev</i>	<i>šav</i>	<i>šev</i>
11	* <i>lstō</i>	cunning	<i>lest</i>	<i>last^a</i>	<i>lešť</i>
12	* <i>dōždžō</i>	rain	<i>děšť</i>	<i>dažd</i>	<i>dážď</i>
13	* <i>ogŕō</i>	fire	<i>oheň</i>	<i>oganj</i>	<i>oheň</i>
14	* <i>dōska</i>	board	<i>deska</i>	<i>daska</i>	<i>doska</i>
15	* <i>bōzō</i>	elder tree	<i>bez</i>	<i>baz</i>	<i>baza</i>
16	* <i>pōstrō</i>	variegated	<i>pestrý</i>	— ^b	<i>pestrý</i>
17	* <i>stōblo</i>	stalk	<i>stěblo</i>	<i>stablo</i>	<i>steblo</i>
18	* <i>donōsō</i>	today	<i>dnes</i>	<i>danas</i>	<i>dnes</i>
19	* <i>pōnō</i>	stump	<i>peň</i>	<i>panj</i>	<i>peň</i>
20	* <i>zōtō</i>	son-in-law	<i>zeť</i>	<i>zet^c</i>	<i>zať</i>
21	* <i>tōnōkō</i>	thin	<i>tenký</i>	<i>tanak</i>	<i>tenký</i>
22	* <i>lgōkō</i>	light	<i>lehký</i>	<i>lak</i>	<i>lahký</i>
23	* <i>jōgōla</i>	needle	<i>jehla</i>	<i>igla^d</i>	<i>ihla</i>
Strong jer vocalized:			23/23	22/22	23/23

^aArchaic/literary; replaced by *lukavstvo* in colloquial Serbian. ^bRoot lost in standard Serbian; replaced by *šaren*. ^cSerbian /e/ instead of expected /a/; the jer still vocalized, but with irregular quality. ^dInitial **jō*- regularly yields Serbian /i/ (cf. **jōmę* → *ime* ‘name’).

liquid-adjacent strong jer that vocalized to /e/ would suffice to falsify the categorical claim; none has been found.

3.2 Category B: Strong jers adjacent to liquids

Table 2 presents strong-position jers adjacent to /r/ (from PIE syllabic **r*), and Table 3 those adjacent to /l/ (from PIE syllabic **l*).

Table 2: Strong-position jers adjacent to /r/: Czech outcomes.

#	Proto-Slavic	Gloss	PIE	Czech	→ /e/?	Outcome
1	*sǝmǝrtǝ	death	*mǝrtis	smrt	no	syll. r
2	*mǝrtǝvǝ	dead	*mǝrtwós	mrtvý	no	syll. r
3	*vǝrxǝ	top	*wǝrs-	vrch	no	syll. r
4	*tǝrgǝ	market	—	trh	no	syll. r
5	*kǝrkǝ	neck	*(s)ker-	krk	no	syll. r
6	*pǝrstǝ	finger	*pǝrsth ₂	prst	no	syll. r
7	*tǝrnǝ	thorn	*tǝrnós	trn	no	syll. r
8	*pǝrvǝ	first	*pǝrh ₂ wós	prvý	no	syll. r
9	*gǝrdlo	throat	*g ^w ǝrh ₃ d ^h lom	hrdlo	no	syll. r
10	*zǝrno	grain	*gǝrh ₂ nóm	zrno	no	syll. r
11	*sǝrdǝce	heart	*kǝrd-	srdce	no	syll. r
12	*čǝrvǝ	worm	*k ^w ǝrmis	červ	SEC.	OldCz. črv
13	*čǝrnǝ	black	*kǝrsnós	černý	SEC.	OldCz. črný
14	*čǝrvenǝ	red	← *čǝrvǝ	červený	SEC.	OldCz. črvený
15	*čǝrstǝvǝ	stale	—	čerstvý	SEC.	OldCz. črstvý
16	*bǝrvǝno	beam	*b ^h ǝrw-	břevno	SEC.	epenthesis
17	*čǝrta	line	*(s)kǝrt-	črt / čára	mixed	črt = syll. r

Table 3: Strong-position jers adjacent to /l/: Czech outcomes.

#	Proto-Slavic	Gloss	PIE	Czech	→ /e/?	Outcome
18	*vǝlkǝ	wolf	*wǝlk ^w os	vlk	no	syll. l
19	*pǝlnǝ	full	*pǝlh ₁ nos	plný	no	syll. l
20	*vǝlna	wool	*h ₂ wǝlh ₁ neh ₂	vlna	no	syll. l
21	*slǝza	tear	—	slza	no	syll. l
22	*žǝltǝ	yellow	*g ^h ǝlh ₃ tos	žlutý	no → /u/	hard l → lu
23	*dǝlgǝ	long	*dǝlh ₁ g ^h os	dlouhý	no → /u/	hard l → lu → ou
24	*dǝlgǝ	debt	*d ^h ǝlg ^h -	dluh	no → /u/	hard l → lu
25	*tǝlstǝ	thick	—	tlustý	no → /u/	hard l → lu
26	*blǝxa	flea	—	blecha	SEC.	OldCz. blcha
27	*plǝtǝ	flesh	*pleth ₂ -	pleť	SEC.	OldCz. plt

Results for Category B:

- Syllabic consonant preserved (no vocalization at all): **15 words**
- Hard syllabic *l* → /u/ (not /e/): **4 words**
- Secondary epenthesis (13th–15th c., from Old Czech syllabic consonant): **7 words**
- Direct vocalization to /e/ by Havlík’s law: **0 words**

The vocalization rate for liquid-adjacent strong jers is **0%** ($n = 27$).

3.3 The contingency table

Table 4: Havlík’s law outcomes by environment (Czech, strong-position jers only).

	Vocalized to /e/	Did not vocalize to /e/
Non-liquid environment	23	0
Liquid-adjacent environment	0	27

Fisher’s exact test on the 2×2 table yields $p < 10^{-14}$. The asymmetry is not a statistical tendency; it is categorical. The same split is replicated independently in Serbian (22/22 vs. 0) and Slovak (23/23 vs. 0).

4 The secondary epenthesis argument

Seven Category B words show /e/ in Modern Czech: *červ* ‘worm’, *černý* ‘black’, *červený* ‘red’, *čerstvý* ‘stale’, *břevno* ‘beam’, *blecha* ‘flea’, and *pleť* ‘complexion’. A reviewer might object that these represent Havlík’s law vocalization and thus constitute counterexamples. Three lines of evidence demonstrate that they do not.

4.1 Old Czech attestation

Historical texts from the 13th–14th centuries attest Old Czech forms **without** /e/: *črv* (not *červ*), *črný* (not *černý*), *črvený* (not *červený*), *črstvý* (not *čerstvý*), *blcha* (not *blecha*), *plt* (not *pleť*), *stblo* (not *stéblo*). These forms are attested in 13th–14th century sources including the *Dalimil Chronicle* (ca. 1314) and other early Old Czech texts (see Lamprecht 1987: 104–112; Komárek 1969 for a systematic overview). The syllabic consonant stage is directly documented in the written record. The /e/ appeared later.

4.2 Chronological separation

Havlík’s law operated during the 10th–12th centuries—the period of the “fall of the jers” across Slavic. The secondary epenthesis that produced *črv* → *červ* and *blcha* → *blecha*

occurred in the **13th–15th centuries**, well after the jer fall was complete. These are two distinct sound changes separated by centuries. The /e/ in *černý* was never a jer; it is a later innovation to resolve consonant clusters that had become too complex.

4.3 The /u/ test

When hard syllabic /l/ resolved in Czech, the epenthetic vowel was /u/, not /e/:

<i>*žltō</i>	→ <i>žlutý</i>	(not <i>*žletý</i>)
<i>*d̥lgō</i>	→ <i>dlouhý</i>	(via <i>dlúhý</i> , not <i>*dlehý</i>)
<i>*d̥lgō</i>	→ <i>dluh</i>	(not <i>*dleh</i>)
<i>*t̥lstō</i>	→ <i>tlustý</i>	(not <i>*tlestý</i>)

If Havlík’s law had vocalized these strong jers, the output should have been /e/—as it is in 100% of Category A cases. The fact that /u/ appears instead proves that a different process was at work: resolution of a syllabic consonant, not vocalization of a jer. The vowel quality tracks the phonetic character of the liquid (back /u/ for velarized /l/), not the quality predicted by jer vocalization.

5 Cross-Slavic evidence

The asymmetry documented in Czech is not a Czech-internal peculiarity. Other Slavic branches show parallel patterns, each in their own way.

5.1 Serbian

Serbian vocalizes non-liquid strong jers to /a/ consistently: **s̥nō* → *san*, **d̥nō* → *dan*, **p̥s̥ō* → *pas*. For liquid-adjacent jers, the picture is different:

- **Syllabic /r/ is preserved** across a broad inventory: *smrt* ‘death’, *prst* ‘finger’, *crn* ‘black’, *trn* ‘thorn’, *trg* ‘market’, *vrh* ‘top’, *krv* ‘blood’, *srp* ‘sickle’, *grb* ‘hump’, *brz* ‘fast’, *grlo* ‘throat’, *zrno* ‘grain’, *srce* ‘heart’, *drvo* ‘wood’, *tvrđ* ‘hard’, *crv* ‘worm’, *prvi* ‘first’—at least 20 common words, all from PIE syllabic **r̥* or Proto-Slavic *ьr* positions.
- **Syllabic /l/ resolved to /u/**: *vuk* ‘wolf’ (< *vlk*), *vuna* ‘wool’ (< *vl̥na*), *pun* ‘full’ (< *pl̥n*), *žut* ‘yellow’ (< *lt*).

The vowel inserted for **l̥* positions is /u/, not /a/. If Havlík’s law had vocalized these jers, we would expect /a/—the same reflex as all Category A jers. The /u/ points to a different process: resolution of a syllabic consonant.

A further piece of evidence comes from the 19th-century Serbian spelling reform. When Vuk Karadžić removed jers from the orthography, words like трънъ became *trn* with **no change in pronunciation**. The /r/ did not “become” syllabic at that point—it already was. The jer had been an orthographic fossil for centuries.

The **asymmetry between /r/ and /l/** is itself informative. Under the standard model, both positions contained “the same jer”: why would they behave differently? Under our model, both were syllabic consonants; syllabic /r/ was more typologically stable than syllabic /l/ and survived longer, while /l/ independently resolved to /u/ (completed by ca. 1400, based on manuscript evidence).

5.2 Russian: The polnoglasie connection

Russian presents an initially challenging case for our argument. East Slavic vocalized **all** jers in liquid contexts to full vowels: **vьlkъ* → волк [volk], **sъmьrtь* → смерть [smʲertʲ], **pьrstъ* → перст [pʲerst]. This appears to support the standard model. However, closer examination of the vowel quality reveals that the Russian data *support* rather than contradict our proposal.

5.2.1 The vowel quality anomaly

Under Havlík’s law, front jer ь vocalizes to /e/ and back jer ъ vocalizes to /o/. In non-liquid contexts, Russian follows this pattern consistently: **dьnъ* → день /e/, **sъnъ* → сон /o/. In liquid contexts, the prediction holds for ь before /r/: **vьrxъ* → верх /e/, **pьrstъ* → перст /e/, **sъmьrtь* → смерть /e/, **zьrno* → зерно /e/, **vьrba* → верба /e/.

But for ь before /l/, the prediction **fails**:

Proto-Slavic	Jer type	Havlík predicts	Russian	Actual vowel	Match?
<i>*vьlkъ</i>	ь (front)	/e/ (*велк)	волк	/o/	no
<i>*pьlnъ</i>	ь (front)	/e/ (*пельный)	полный	/o/	no
<i>*žьltъ</i>	ь (front)	/e/ (*желтый)	жёлтый	/o/ (via ě)	no

In every case, front jer ь before /l/ yields Russian /o/ instead of the expected /e/. The vowel quality is determined not by the jer color but by the **liquid consonant**: velarized (dark) /l/ conditions backing and rounding of the adjacent vowel, pulling it from the /e/ region to /o/.

5.2.2 The polnoglasie parallel

This vowel-quality mismatch is not an isolated anomaly. It follows a pattern already well known in East Slavic historical phonology: **polnoglasie** (pleophony; Shevelov 1964:

421–446). East Slavic resolved Proto-Slavic CRC clusters by inserting (copying) a vowel after the liquid, with velarized /l/ conditioning backing of the copied vowel to /o/:

<i>*gordz</i> (TORT)	→ город (ToroT)	vowel copy after liquid
<i>*golva</i> (TolT)	→ голова (ToloT)	vowel copy after liquid
<i>*bergz</i> (TerT)	→ берег (TereT)	vowel copy after liquid
<i>*melko</i> (TelT)	→ молоко (ToloT)	backing before /l/

The last example is critical: **TelT* → ToloT. The original /e/ in **melko* was backed to /o/ before /l/—the **exact same process** that turned the expected /e/ from ь into /o/ in волк, полный, долг.

The structural parallel is systematic:

Pattern	Input	East Slavic output	Process
Classic polnoglasie	TorT	ToroT	vowel copy
Liquid-jer (ь+r)	TьrT	TerT	/e/ from ь
Liquid-jer (ь+l)	TьlT	TolT	/o/ — matches polnoglasie

5.2.3 The geographic isogloss

The geographic split between South Slavic and East Slavic is **identical** for both TORT and TьRT words:

	South Slavic	East Slavic
<i>*gordz</i> (TORT)	градъ (metathesis)	город (polnoglasie)
<i>*golva</i> (TolT)	глава (metathesis)	голова (polnoglasie)
<i>*vɔlkz</i> (TьlT)	влькъ > Serb. <i>vuk</i>	волк (/o/ epenthesis)
<i>*szmьrtb</i> (TьrT)	Serb. <i>smrt</i> (syll. <i>r</i>)	смерть (/e/ epenthesis)
<i>*pьrstz</i> (TьrT)	Serb. <i>prst</i> (syll. <i>r</i>)	перст (/e/ epenthesis)

In every case, South Slavic preserves or reconfigures the consonant cluster, while East Slavic breaks it up with a full vowel. This is the same phonotactic constraint operating on two successive waves of CRC clusters: the TORT sequences (8th–9th c.) and the TьRT sequences exposed as CRC clusters after the jer fall (10th–12th c.).

5.2.4 Reinterpretation

We propose that what is traditionally described as “vocalization of liquid-context jers” in East Slavic is better analyzed as **polnoglasie applied to syllabic consonants**. East Slavic had a productive phonotactic constraint against CRC sequences; when the jer fall exposed syllabic consonants as CRC clusters, the same epenthetic mechanism

that had resolved TORT sequences was applied to T_lT and T_rT sequences. The vowel quality was determined by the liquid (backing before /l/, preservation before /r/), not by a jer that had already ceased to exist.

Under this analysis, Russian does not constitute evidence *for* the standard model. The East Slavic data are compatible with our proposal: the jers in liquid contexts were non-vocalic, and their apparent “vocalization” was in fact independent epenthesis—the same process, by a different name, that every other IE branch also applied to PIE syllabic resonants.

Further support comes from the Old Novgorod dialect (Zaliznyak 2004), which shows T_lR_lT / T_lR_lT patterns—a **copy** of the jer inserted after the liquid, structurally identical to how polnoglasie copies the full vowel. In Novgorod, the two jers apparently *neutralized* in this position, and the resulting vowel quality was conditioned by the consonantal environment rather than the original jer color.

5.3 The cross-Slavic summary

Table 5: Cross-Slavic treatment of liquid-adjacent jers vs. non-liquid jers.

Branch	Non-liquid jer →	Liquid jer →	Same process?
Czech/Slovak	/e/	syllabic C	no
Serbian	/a/	syll. <i>r</i> (kept) / /u/ (<i>l</i>)	no
Russian	/o/ or /e/	/o/ or /e/ (polnoglasie-like)	no (see §5.2)
Bulgarian	/e/ or /o/	/ɤ/ [ɣ] + liquid	no

In all four branches examined in detail, liquid-adjacent jers **behave differently** from non-liquid jers.⁴ For Czech and Serbian the distinction is categorical; for Russian, the vowel quality in liquid-adjacent positions follows polnoglasie patterns rather than standard jer vocalization (§5.2). Bulgarian is particularly informative. Non-liquid front jer ь regularly yields /e/ (ден < **dnъ*, пер < **prъsъ*). If Havlík’s law had applied to liquid-adjacent jers, front jer ь before a liquid should likewise yield /e/. It does not:

⁴Polish also shows distinct treatment of liquid-adjacent jers, with variable outcomes (/i/, /e/, /a/) contrasting with uniform /e/ for non-liquid jers (Rubach 1993). However, the Polish situation is complicated by dialect variation and morphological conditioning, and we defer a full treatment to future work.

Proto-Slavic	Jer	Gloss	Havlík predicts	Bulgarian	Actual	Match?
* <i>dnъ</i>	ь	day	/e/	ден	/e/	yes (non-liquid)
* <i>psъ</i>	ь	dog	/e/	пес	/e/	yes (non-liquid)
* <i>vьlkъ</i>	ь	wolf	/e/	вълк	/ъ/ [ɤ]	no
* <i>pьlnъ</i>	ь	full	/e/	пълн	/ъ/ [ɤ]	no
* <i>žьltъ</i>	ь	yellow	/e/	жълт	/ъ/ [ɤ]	no
* <i>pьrstъ</i>	ь	finger	/e/	пръст	/ъ/ [ɤ]	no
* <i>tьrnъ</i>	ь	thorn	/e/	трън	/ъ/ [ɤ]	no

Front jer ь yields /e/ in non-liquid contexts but /ъ/ [ɤ] before liquids—a back vowel where a front vowel is predicted (see Scatton 1984 for the Bulgarian jer reflexes). This parallels Russian exactly (§5.2), where front jer before /l/ yields /o/ instead of /e/. In both branches, the vowel quality at liquid-adjacent positions is determined by the consonantal environment, not by the jer color—consistent with epenthesis into a syllabic consonant rather than vocalization of a jer.

6 Engagement with the standard account

6.1 The implicit exception

The standard account of Havlík’s law does not contain an explicit exception clause for liquid contexts. Instead, textbooks present jer vocalization and syllabic consonant formation as complementary outcomes of the same process, without flagging the logical tension. A representative formulation might read: “Strong jers vocalized; in liquid environments, the liquid became syllabic.” But this formulation masks a genuine asymmetry: *why* does the liquid “become syllabic” instead of the jer vocalizing, as it does in every non-liquid context?

The standard answer, insofar as one is given, relies on rule ordering: the jer weakens, the liquid—being the most sonorous adjacent segment—fills the vacated nuclear position. But this is precisely the kind of post hoc explanation that our alternative renders unnecessary.

6.2 Scheer’s structural analysis

Scheer’s (2003, 2004) CVCV analysis of Czech syllabic consonants provides the most detailed structural account of liquid-jer interactions in the literature. In his Government Phonology framework, syllable structure is represented as a strict alternation of consonantal (C) and vocalic (V) positions. A syllabic consonant is analyzed as a liquid that spreads into an adjacent empty vocalic position, acquiring nuclear properties (stress-bearing, mora-contributing). Scheer distinguishes two structural types in Czech:

1. **Syllabic consonants** (from CS **t_{br}t*, jer before liquid): the jer is lost, leaving an empty nucleus which the liquid fills through leftward spreading. Example: **v_{bl}kz* → *vlk*, where /l/ spreads into the nuclear position originally marked by Ъ.
2. **Trapped consonants** (from CS **tr_{br}t*, jer after liquid): the jer is lost, and the liquid is “trapped” between two consonants, becoming syllabic through government relations. Example: **b_{br}ati* → *brát*, where /r/ is trapped and syllabifies.

Scheer’s analysis is **structurally compatible with both the standard and alternative models**. The key point is that the synchronic phonological structure—a liquid associated with a nuclear position—is *identical* under both interpretations:

- **Standard model:** The nuclear position was occupied by a real reduced vowel (PBS **i*), which later deleted, leaving an empty nucleus that the liquid filled.
- **Alternative model (ours):** The nuclear position was *always* empty (or minimally vocalic), with the liquid functioning as the nucleus from the start; the jer was a scribal convention marking this syllabic function.

In both cases, the *outcome* is a liquid in a nuclear position. The difference is purely diachronic: was there an intermediate stage with a full vowel, or was the liquid quasi-syllabic throughout?

Scheer’s framework does not adjudicate this question because it is designed to model synchronic phonological structure, not historical phonology. His analysis takes the OCS spelling (jer + liquid) as a starting point and derives the Modern Czech outcome (syllabic consonant) through independently motivated phonological operations (nuclear deletion, segmental spreading, government). The framework is neutral on whether the OCS jer notation represented a real vowel or a notational device.

What Scheer’s analysis contributes to our argument: Two key insights from Scheer support the broader claim that OCS jer notation was functionally heterogeneous:

1. **Structural heterogeneity of jers.** Scheer shows that syllabic consonants (**t_{br}t*) and trapped consonants (**tr_{br}t*) are **structurally distinct** categories in Modern Czech, even though both were written with jer in OCS. This demonstrates that OCS jer notation did not represent a uniform phonological category—what looks identical orthographically (Ь) corresponds to different underlying structures. Our proposal extends this insight: not only do jers in different *positions* (before vs. after liquids) correspond to different structures, but jers in liquid contexts vs. non-liquid contexts correspond to fundamentally different *types* of segment (notational vs. real vowel).

2. **Independent evidence for empty nuclei.** Scheer’s framework requires that certain positions contain **empty nuclei**—vocalic positions in the syllable structure that have no segmental content. Under the standard model, these empty nuclei arise *after* the jer is deleted. Under our model, they were empty (or quasi-empty) *from the start*. Either way, the existence of structurally projected but phonetically empty vocalic positions is independently required by Modern Czech phonotactics. Our proposal simply pushes the origin of these empty nuclei earlier in the historical timeline.

What our data adds: Scheer’s analysis does not explain—or attempt to explain—the categorical 100% vs. 0% vocalization asymmetry documented in §3. His framework derives the *outcome* (syllabic consonants) from the *input* (OCS jer + liquid), but it is silent on why Havlík’s law vocalization operates categorically differently in liquid vs. non-liquid environments. In Scheer’s terms, one could ask: why does the strong-position nuclear slot vocalize to /e/ in **sɔnɔ* → *sen* but remain empty (to be filled by liquid spreading) in **vɔlkɔ* → *vlk*? The CVCV framework has no mechanism to predict this split.

Our answer is historical: the nuclear slot in **sɔnɔ* was occupied by a real vowel (PIE **u*), which vocalizes under Havlík’s law; the nuclear slot in **vɔlkɔ* was never occupied by a real vowel (PIE **l* was quasi-syllabic), so there was nothing to vocalize. The asymmetry arises from a difference in *etymological input*, not synchronic phonological structure.

A note on trapped consonants is in order. Our argument has focused on the **tɔrt* type (jer before liquid), where the strong-position jer fails to vocalize. What about the **trɔt* type (jer after liquid), where Scheer posits a structurally distinct “trapped” outcome? Under our model, both jer positions are notational: neither the jer in **vɔlkɔ* nor the jer in **bɔrati* represents a real reduced vowel. The prediction is the same in both cases—the jer should not participate in Havlík’s law—and the Czech outcomes confirm this:

Proto-Slavic	Type	Czech	Outcome	Jer vocalized?
<i>*bɔrati</i>	<i>*trɔt</i>	<i>brát</i>	trapped <i>r</i>	no (á < root vowel)
<i>*krɔstɔ</i>	<i>*trɔt</i>	<i>kříž</i>	trapped <i>r</i>	no
<i>*drɔva</i>	<i>*trɔt</i>	<i>dříví</i>	trapped <i>r</i>	no
<i>*blɔstěti</i>	<i>*tlɔt</i>	<i>blýskat</i>	trapped <i>l</i>	no

In no case does the jer in a trapped-consonant position vocalize to /e/. We note, however, that these data are *neutral* between the two models: Scheer’s standard CVCV analysis independently predicts that trapped-consonant jers do not vocalize (the liquid

spreads into the vacated nuclear position before vocalization can apply). The trapped-consonant cases are therefore compatible with our proposal but do not constitute independent evidence for it. The distinguishing evidence comes from the **tbrt* type (§3), where the standard model predicts vocalization and our model does not. Scheer’s structural distinction between syllabic and trapped consonants remains valid as a synchronic description; our proposal adds the diachronic claim that neither type ever contained a real vowel at the jer position.

In summary, Scheer’s structural analysis and our diachronic proposal are complementary. Scheer provides the synchronic phonological machinery to derive syllabic consonants from underlying representations; we provide the historical explanation for why the underlying representations differ categorically between liquid and non-liquid environments. Together, the two accounts form a more complete picture than either alone.

6.3 The ɶ/ɤ distinction

One argument for the standard model is that the front/back jer distinction in liquid contexts (ɶr vs. ɤr) implies a vocalic element with quality. A purely syllabic consonant has no front/back dimension.

A systematic survey of 74 Proto-Slavic words with jer adjacent to a liquid (*r* or *l*) substantially weakens this objection. The jer color in these positions turns out to be **etymologically determined**, with near-categorical accuracy (Table 6): front jer ɶ correlates almost perfectly with PIE syllabic resonant origin, while back jer ɤ correlates perfectly with non-syllabic-resonant sources (loanwords, laryngeal sequences, substrate words). The minimal pair **dɔlgɔ* ‘long’ (from PIE **dl̥h₁gʰos*, front jer) vs. **dɔlgɔ* ‘debt’ (from Gothic, back jer) epitomizes the pattern: same consonant skeleton, opposite jer color, opposite etymological origin.

Table 6: Jer color and etymological source in liquid contexts ($n = 74$).

	PIE <i>*r̥</i> / <i>*l̥</i>	Other source	Total
ɶ + liquid	35 (94.6%)	2 (5.4%)	37
ɤ + liquid	0 (0%)	37 (100%)	37

Representative examples illustrate the pattern:

Proto-Slavic	Jer	Gloss	Source	PIE syll. resonant?
* <i>volkǫ</i>	Ь	wolf	PIE * <i>wlk^wos</i>	yes
* <i>pǫrstǫ</i>	Ь	finger	PIE * <i>prsth₂</i>	yes
* <i>čǫrnǫ</i>	Ь	black	PIE * <i>kṛsnós</i>	yes
* <i>zǫrno</i>	Ь	grain	PIE * <i>grh₂nóm</i>	yes
* <i>dǫlgǫ</i>	Ь	long	PIE * <i>dḷh₁g^hos</i>	yes
* <i>volna</i>	Ь	wool	PIE * <i>h₂wḷh₁neh₂</i>	yes
* <i>dǫlgǫ</i>	Ѣ	debt	Gothic <i>dulgs</i>	no (loanword)
* <i>xǫlmǫ</i>	Ѣ	hill	PGmc * <i>hulmaz</i>	no (loanword)
* <i>gǫrdlo</i>	Ѣ	throat	PIE * <i>g^wṛh₃d^hlom</i>	no (laryngeal)
* <i>kǫrmiti</i>	Ѣ	to feed	PIE * <i>k^wṛHm-</i>	no (laryngeal)
* <i>tǫrgǫ</i>	Ѣ	market	substrate	no (substrate)
* <i>tǫlstǫ</i>	Ѣ	thick	unknown	no (unknown)

This distribution shows that jer color in liquid contexts encodes inherited PBS-level etymological information (corresponding to the PBS **iR*/**uR* split; Mayer 1991), not independent Slavic vocalic quality—exactly what one would expect of a notational system preserving traditional spelling distinctions from an earlier stage, not of a phonologically active vowel whose quality is determined by its synchronic environment.⁵

6.4 The Balto-Slavic agreement

Baltic and Slavic agree on front (**i*) vs. back (**u*) quality in the same liquid-context words (~73% **i*; Mayer 1991), suggesting a shared PBS innovation rather than independent development. This is genuine evidence for a PBS stage with *some* vocalic quality in these positions. We do not dismiss this evidence.⁶

However, the agreement is equally compatible with a PBS stage where the resonant was quasi-syllabic with incipient vocalic coloring—not a full vowel. Baltic developed this coloring into a full vowel **i* (Lithuanian *vilkas*, *vilna*, *pilnas*); Slavic preserved the syllabic consonant with the inherited coloring encoded in the Ѣ/Ь spelling (see §6.3). The agreement on front/back quality requires a shared starting point, but that starting point need not have been a full short vowel.

The two models make a concrete, distinguishable prediction. If the PBS stage contained a real short vowel, that vowel should participate in Havlík’s law like all other real reduced vowels—that is, it should vocalize in strong position. The data in §3 show that

⁵The full 74-word corpus, conditioning environments, and diagnostic applications of the jer-color pattern are developed in a separate study.

⁶Kortlandt (1989) proposes a detailed PBS chronology in which syllabic resonants acquired epenthetic vowels at a relatively late stage, after other PBS developments. Our proposal is compatible with a late or minimal PBS vocalization; the key question is whether that vocalization produced a full short vowel that should have participated in Havlík’s law, or a quasi-syllabic segment that did not.

it categorically does not. This is the empirical test that distinguishes “quasi-syllabic consonant with incipient coloring” from “real short vowel”: a real vowel vocalizes; liquid-context jers never do.

A clarification of terminology is warranted here. When we call liquid-context jers “notational,” we do not mean that they encoded nothing real. The ɤ/ɤ distinction in these positions faithfully recorded a genuine phonetic property—the front or back coloring of the syllabic resonant, inherited from PBS. What we mean is that this coloring was a *secondary property of the consonant*, not an independent vocalic segment. The jer grapheme represented the consonant’s coloring, much as a diacritic represents a secondary articulation, rather than a separate vowel occupying its own position in the syllable. The critical difference is phonological status: a full short vowel participates in vowel-specific processes (Havlík’s law vocalization); a colored consonant does not. The 68/68 vs. 0/27 split is the empirical reflex of this distinction.

Furthermore, the Balto-Slavic agreement on quality distribution is itself informative for our argument. Mayer’s (1991) comprehensive study of 215 PBS lexical items shows that the front (**i*) reflex accounts for ~73% of cases, back (**u*) for ~17%, and both for ~10%. If the PBS vowel had been a robust, phonologically active short vowel, we would expect the conditioning of its front/back quality to be well understood. In fact, over a century of scholarship has failed to establish the conditioning definitively (proposals by Vaillant 1950, Kuryłowicz 1956, Endzelīns 1971, and Stang 1957 all capture only a subset of cases). This opacity is more consistent with a quasi-syllabic resonant whose incipient coloring was phonetically variable than with a firmly established short vowel.

7 The proposal

7.1 Two types of jer

We propose that the OCS graphemes ɤ and ɤ served at least three distinct functions, two of which are phonologically genuine and one of which is notational:

Table 7: Three functions of the jer graphemes.

Function	Example	PIE source	Real vowel?
Reduced vowel (< PIE <i>*u</i>)	сънъ ‘sleep’	<i>*supnos</i>	yes
Reduced vowel (< PIE <i>*i</i>)	дънь ‘day’	<i>*dyew-</i>	yes
Syllabic C notation	влькъ ‘wolf’	<i>*wlk^wos</i>	no
Loanword epenthesis	стъкло ‘glass’	Gmc <i>*stikls</i>	no

A fourth function—loanword epenthesis—is included for completeness: words like стъкло ‘glass’ (from Germanic **stikls*) acquired a jer in Slavic to break up consonant clusters foreign to Slavic phonotactics. These epenthetic jers, like syllabic-consonant

jers, were not inherited reduced vowels and similarly fall outside the domain of Havlík’s law.⁷

Havlík’s law applies to the first two categories: genuine reduced vowels that alternate between vocalization (strong position) and deletion (weak position). It does not apply to the third or fourth categories, because there is no inherited vowel to vocalize. The 100% vs. 0% split in the Czech data (Tables 1–3) reflects this categorical distinction.

An alternative interpretation should be considered and rejected. One might propose that liquid-context jers *were* real vowels but ultra-short ones—phonetically present but already below the threshold for Havlík’s law vocalization. On this view, the 0/27 result reflects phonological inertness, not absence. Five arguments weigh against this alternative:

1. **Parsimony.** The “inert vowel” hypothesis requires positing a special phonological category—vowels too short to vocalize yet real enough to carry front/back quality—that has no independent motivation in Slavic. We would need to explain why *only* liquid-adjacent jers fell into this category while all other jers, including those in complex clusters (**p̣str̥* → *pestrý*, **d̥žd̥ẓ̌* → *děšť*), vocalized normally.
2. **The /u/ test.** When hard syllabic /l/ resolved in Czech, it produced /u/ (*žlutý*, *dlouhý*)—the same outcome regardless of whether a jer was once present. If liquid-context jers had been real vowels, however short, we would expect some interaction with the resolution process. Instead, the vowel quality tracks the liquid consonant, exactly as in positions where no vowel was ever posited.
3. **The polnoglasie parallel.** Russian resolves liquid-context jers by the same epenthetic mechanism it applies to TORT sequences (§5.2), where no jer was ever present. If liquid-context jers had been real vowels, there would be no reason for Russian to treat them identically to vowelless consonant clusters.
4. **Cross-IE and Slavic-internal divergence.** Every IE branch independently inserts a *different* vowel at PIE syllabic resonant positions. Within Slavic, five branches produce five different vowels at the same position (Czech *vlk*, Serbian *vuk*, Russian волк, Polish *wilk*, Bulgarian вълк). If a real vowel had been inherited, the question arises why five branches would each independently replace it. Independent epenthesis into an empty position requires no such explanation.
5. **The jer-color diagnostic.** If liquid-context jers were real vowels, their front/back quality should reflect Slavic-internal phonological conditioning. Instead, it reflects PIE etymology with near-categorical accuracy (Table 6: 94.6% of ʁ+liquid words

⁷Compare Czech *sklo* (jer deleted, no vocalization) vs. *sen* < **s̥n̥s̥* (jer vocalized). The loanword jer in **sṭklo* behaves like a liquid-context jer, not like a genuine reduced vowel.

trace to PIE syllabic resonants; 0% of ɣ+liquid words do). This distribution is expected from a notation preserving inherited distinctions, not from a phonologically active segment whose quality is determined by its synchronic environment.

The cumulative weight of these arguments favors the stronger claim: liquid-context jers were not merely inert vowels but notational representations of syllabic consonants that never had independent vocalic status in Slavic.

7.2 A reformulated Havlík’s law

Under the standard formulation, Havlík’s law requires an implicit exception: “Strong jers vocalize—*except in liquid environments, where the liquid becomes syllabic instead.*” Under our reformulation, no exception is needed:

Havlík’s law (reformulated): In Proto-Slavic words containing genuine reduced vowels (ɣ from **u*, ɤ from **i*), weak-position jers are deleted and strong-position jers vocalize to full vowels. Orthographic jers representing syllabic consonants (from PIE **l*, **r*) are outside the domain of the law.

This reformulation makes Havlík’s law **more regular**, not less. The apparent exception for liquid contexts vanishes because the jers in those positions were never part of the law’s input.

7.3 Implications for Proto-Balto-Slavic

If liquid-context jers were not real vowels, the standard PBS reconstruction of syllabic resonants via an intermediate **iR*/**uR* stage requires re-examination. Under the standard model, Slavic first epenthesized a vowel (PIE **wlk^wos* → PBS **wilkas*) and then reduced it back to a syllabic consonant—two changes that cancel each other out. Our data raise the question of whether the intermediate full-vowel stage existed in Slavic at all. We defer a full treatment of this question to a separate study; the Havlík’s law asymmetry documented here is one empirical argument that such a study must address.

8 Open questions

We flag two questions that require further research:

1. **Non-PIE-syllabic-resonant liquid jers across Slavic.** The back-jer (ɣ) + liquid words identified in Table 6 include loanwords (**dɔlgɔ* ‘debt’ from Gothic, **xɔlmɔ* ‘hill’ and **pɔlkɔ* ‘regiment’ from Germanic), laryngeal sequences (**kɔrkɔ*

‘neck’, **gǝrdlo* ‘throat’), and substrate words. These do not derive from PIE syllabic resonants, yet their Czech outcomes pattern identically with Category B: *dľuh*, *chľum*, *plľuk* (hard l → /u/); *krľk*, *hrľlo* (syllabic r). None shows /e/ vocalization. This suggests that the Category A/B split tracks *phonological context* (liquid adjacency), not just PIE etymological class. A fuller investigation of these cases across Serbian and Slovak would further test this generalization.

2. **Old Russian manuscripts.** Is there evidence for a syllabic consonant stage in Old Russian *before* full vocalization? If East Slavic manuscripts show variation between jer + liquid and liquid-only spellings (as OCS manuscripts do), this would support a syllabic consonant intermediate stage.

9 Conclusion

We have documented a categorical asymmetry in the outcomes of Havlík’s law across Slavic: strong-position jers vocalize to full vowels in 100% of non-liquid cases—Czech /e/ ($n = 23$), Serbian /a/ ($n = 22$), Slovak /e/ ($n = 23$)—but in 0% of liquid-adjacent cases ($n = 27$). Across three independent branches, 68 out of 68 non-liquid strong jers vocalized while zero liquid-adjacent strong jers did. The seven Modern Czech words with /e/ near liquids are secondary epenthesis, documented in Old Czech as syllabic consonant forms, and separated from Havlík’s law by two to three centuries. The resolution of hard syllabic /l/ to /u/ (not /e/) confirms a process distinct from jer vocalization. In Russian, the vowel quality at liquid-jer positions follows polnoglasie patterns (front jer Ъ → /o/ before /l/, matching **TelT* → ToloT) rather than standard jer vocalization, indicating that East Slavic “vocalization” was independent epenthesis to resolve syllabic consonants, not Havlík’s law.

We argue that this asymmetry constitutes evidence that liquid-context jers in OCS were notational conventions for syllabic consonants, not genuine reduced vowels. Havlík’s law, reformulated as applying only to real reduced vowels (from PIE **u/ *i*), becomes more regular: the “liquid exception” disappears because there was never a vowel to except.

The argument has implications for Proto-Balto-Slavic reconstruction: if liquid-context jers were never real vowels, the standard PBS **iR/ *uR* intermediate stage requires re-examination—a question we defer to a separate study.

Our proposal does not invalidate Havlík’s law. It refines its domain: the law correctly describes the behavior of genuine reduced vowels, which is exactly the category it was designed to explain. What it does not describe is the behavior of positions that never had a vowel to begin with.

A final observation underscores the point. PIE **wlkʷos* ‘wolf’ yields Czech *vlk*,

Serbian *vuk*, Russian волк, Polish *wilk*, and Bulgarian вълк—five branches, five different vowels at the same position. Under the standard model, a single inherited reduced vowel produced five different outcomes; under ours, each branch independently resolved a syllabic consonant, just as each IE branch did before them (Sanskrit *ṛ*, Greek /a/, Latin /o/, Germanic /u/, Lithuanian /i/; cf. Mayer 1991). The instability is inherent to the phonological position, not to the particular vowel that happened to occupy it.

References

- Beekes, R. S. P. (2011) *Comparative Indo-European Linguistics: An Introduction*. 2nd ed. Amsterdam: John Benjamins.
- Bethin, C. Y. (1998) *Slavic Prosody: Language Change and Phonological Theory*. Cambridge: Cambridge University Press.
- Birnbaum, H. (1976) “Another look at Slavic liquid diphthongs.” *Lingua* 38: 1–26.
- Carlton, T. R. (1991) *Introduction to the Phonological History of the Slavic Languages*. Columbus: Slavica Publishers.
- Derksen, R. (2008) *Etymological Dictionary of the Slavic Inherited Lexicon*. Leiden: Brill.
- Endzelīns, J. (1971) *Comparative Phonology and Morphology of the Baltic Languages*. The Hague: Mouton.
- Fortson, B. W. (2010) *Indo-European Language and Culture: An Introduction*. 2nd ed. Malden, MA: Wiley-Blackwell.
- Gouskova, M. & Becker, M. (2013) “Nonce words show that Russian yer alternations are governed by the grammar.” *Natural Language & Linguistic Theory* 31: 735–765.
- Havlík, A. (1889) “K otázce jerové v staré češtině.” *Časopis pro moderní filologii* 1: 76–82.
- Isačenko, A. V. (1970) “East Slavic morphophonemics and the treatment of the jers in Russian: A revision of Havlik’s law.” *International Journal of Slavic Linguistics and Poetics* 13: 73–124.
- Kenstowicz, M. & Rubach, J. (1987) “The phonology of syllabic nuclei in Slovak.” *Language* 63: 463–497.

- Komárek, M. (1969)** *Historická mluvnice česká, I: Hláskosloví*. 2nd ed. Prague: Státní pedagogické nakladatelství.
- Kortlandt, F. (1989)** “Od praindoevropskog jezika do slovenskog (fonološki razvoj).” *Zbornik za filologiju i lingvistiku* 32(2): 41–58.
- Kuryłowicz, J. (1956)** *L’apophonie en indo-européen*. Wrocław: Zakład Narodowy im. Ossolińskich.
- Lamprecht, A. (1987)** *Praslovanština*. Brno: Univerzita J. E. Purkyně.
- Lunt, H. G. (2001)** *Old Church Slavonic Grammar*. 7th ed. Berlin: Mouton de Gruyter.
- Matasović, R. (2014)** *Slavic Historical Linguistics*. Zagreb: Institut za hrvatski jezik i jezikoslovlje.
- Mayer, H. (1991)** “Reflexes of Indo-European syllabic resonants in Baltic, Slavic, Albanian.” *Lituanus* 37(4).
- Rubach, J. (1993)** *The Lexical Phonology of Slovak*. Oxford: Clarendon Press.
- Scatton, E. A. (1984)** *A Reference Grammar of Modern Bulgarian*. Columbus: Slavica Publishers.
- Scheer, T. (2003)** “Syllabic and trapped consonants in (Western) Slavic.” In *Formal Description of Slavic Languages 5 (FDSL 5)*, 119–132.
- Scheer, T. (2004)** *A Lateral Theory of Phonology: What Is CVCV and Why Should It Be?* Berlin: Mouton de Gruyter.
- Shafirovich, S. (2016)** *Syllabic Liquids in Common Slavic*. Ph.D. dissertation, Purdue University.
- Shevelov, G. Y. (1964)** *A Prehistory of Slavic: The Historical Phonology of Common Slavic*. Heidelberg: Carl Winter.
- Stang, C. S. (1957)** *Slavonic Accentuation*. Oslo: Universitetsforlaget.
- Townsend, C. E. & Janda, L. A. (1996)** *Common and Comparative Slavic: Phonology and Inflection, with Special Attention to Russian, Polish, Czech, Serbo-Croatian, Bulgarian*. Columbus: Slavica Publishers.
- Vaillant, A. (1950)** *Grammaire comparée des langues slaves, vol. I: Phonétique*. Lyon: IAC.

Yoo, S.-M. (2012) “The historical development of tense jers in East Slavic.” *Russian Language Studies* (Русский язык исследования) 11(2): 293–319.

Zaliznyak, A. A. (2004) *Drevnenovgorodskij dialekt*. 2nd ed. Moscow: Jazyki slavjanskoj kul'tury.